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Can Standardized Small-Sided Games Be Used as a Tool to Monitor Post-Match Fatigue in Male Academy Soccer Players?

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Abstract

This study aimed to analyze residual post-match fatigue using time-motion analysis across small-sided games (SSGs), fitness performance, and heart rate variability (HRV) in academy soccer players. Fifteen male academy players (18.3 ± 0.6 years) performed four SSG formats: 3 vs. 3 and 3 vs. 3 + 2 goalkeepers, played on small (25 × 18 m; 75 m^2 per player) and large (36 × 25 m; 150 m^2 per player) areas in a four-week experimental design. Two non-match weeks served as baseline weeks, and two weeks followed an official match. Time-motion during SSGs was monitored via GNSS. Before each training session (24, 72, and 120 h post-match), HRV, countermovement jump (CMJ) height, and maximum isometric force (MIF) of the hamstrings were measured. External load metrics were generally lower in post-match sessions than in non-match weeks; however, their low reliability limited the interpretability of SSG outputs. CMJ height was reduced, MIF was elevated post-match, whereas HRV showed distinct mid-week responses, decreasing in starters but increasing in non-starters. Post-match fatigue was evident in neuromuscular performance but was inconsistently reflected in SSG outputs. CMJ height reliably reflects post-match fatigue in male academy soccer players and should be prioritized when monitoring neuromuscular recovery.

Keywords: athlete monitoring; countermovement jump; neuromuscular fatigue; small-sided games; heart rate variability; youth soccer

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1. Introduction

Post-match fatigue is systematically assessed across professional soccer teams, integrating neuromuscular, physiological, and perceptual measurements to determine player readiness [1]. Monitoring external and internal loads is also strongly recommended to evaluate the stress placed on athletes [2]. External load represents the work done by the player [2], quantified through time-motion activity [3,4], while the player's response as a consequence of these physical demands is known as internal load [2] and quantified by measuring physiological (e.g., heart rate, blood lactate, or oxygen consumption) or psychological (e.g., rating of perceived exertion) responses to exercise [3,4]. Fatigue is a complex, multifaceted phenomenon arising from multiple mechanisms [5]. Therefore,

several tests have been reported to monitor residual fatigue [6–8]. While force-platform-based assessments provide comprehensive neuromuscular force–time characteristics [8], countermovement jump (CMJ) height is a meaningful marker of post-match fatigue [9] due to its sensitivity to post-competition neuromuscular impairments [9]. Furthermore, hamstrings isometric peak force at long muscle lengths can be confidently used to assess hamstrings force-producing capacity after a competitive soccer match [6,8]. Fatigue can also induce changes in autonomic nervous system function [10,11], which can be assessed by analyzing heart rate variability (HRV) [12]. Specifically, the raw root mean square of successive differences (RMSSD) has been used as a sensitive parasympathetic indicator in high-level and young soccer players [13,14], with reductions reflecting incomplete physiological recovery [11]. Consequently, HRV provides complementary autonomic insights alongside mechanical neuromuscular markers, capturing the systemic recovery profile following match play.

Nevertheless, the time required and the need for coaches to train technical, tactical, and physiological stimuli during the majority of soccer-based training [15] often lead to the rejection of physical tests in elite soccer teams. As a potential solution, several studies have analyzed whether standardized small-sided games (SSGs) can serve as a field-based assessment of physical fitness [15–19]. In this regard, the global navigation satellite system (GNSS) technology is currently used to monitor the external load in soccer [3,4]. This method analyzes on-field player performance using multiple time-motion or accelerometry-based variables [20]. In contrast, heart rate responses are commonly used to quantify the internal load [4]. Although acceptable reliability has been reported for variables such as total distance (TD) and mean heart rate (HR_{mean}), evidence regarding their validity remains inconsistent, with large variability in high-intensity actions limiting their usefulness as fitness indicators [15–18].

Beyond fitness assessment, SSGs have also been proposed as a means of monitoring training-induced fatigue. Rowell et al. [21] showed that a standardized SSG (5 vs. 5 + 5; 45 × 36 m) was sensitive to weekly reductions in the ratio of flight time to contact time during a CMJ, with decreases greater than 8% from baseline accompanied by lower accelerometry load during the SSGs. However, interpretation of accelerometry-derived metrics is limited by unit placement, orientation noise, and posture-related effects [22,23]. Likewise, time-motion outcomes across SSG formats show high variability [24,25], especially as the number of players increases [26]. Moreover, various factors may influence the physical responses during SSGs [25,27,28]. Pitch size, number of players, and the inclusion of floaters or goalkeepers (GK) are the main variables to consider when designing SSGs [27,29–31]. Specifically, SSGs conducted in larger areas result in greater workloads than those in smaller areas [32], which elicit more accelerations/decelerations but less distance covered at high speeds [27,32,33]. Moreover, the presence of a GK appears to reduce the player workload on small pitches [29,30] but to increase it on larger pitch areas [33]. As reducing the number of players may increase exercise intensity and reduce tactical constraints [25,28], 3-a-side game formats with larger pitch areas and without GK could provide a useful context for detecting post-match variations in physical condition.

Despite these controversies regarding the use of SSGs to monitor training-induced adaptations, to date, there is limited scientific evidence on their use as a physical assessment when their variability (game constraints) is reduced to test post-match fatigue. Therefore, the aim of the present study was to analyze residual post-match fatigue using time-motion performance across 3-a-side SSG formats varying pitch size and GK presence, and fitness performance during a post-match training week in male academy soccer players. We hypothesized that time-motion performance for SSGs played on a larger pitch without a

GK, and physical performance, would detect residual fatigue after the match, although reliability in SSG parameters would be moderate.

2. Materials and Methods

2.1. Exploratory Design

A quasi-experimental design was used to examine changes in time-motion patterns for four 3-a-side SSGs and physiological and muscular fatigue over the typical training week after an official match in young soccer players. Because GK had different requirements during the matches and SSGs, they were excluded from the study sample and analyses. This study was conducted over four consecutive in-season weeks during the latter stages of the competitive season. The external load was monitored throughout the four weeks to control variation in training load. The SSG format was performed with and without GK and conducted on a small pitch (i.e., ssg and ssg + GK) and a large pitch (i.e., LSG and LSG + GK). A similar training-week schedule was used to analyze variations in time-motion and physical performance, selecting non-match weeks (NMW), with two training weeks without a previous match, and post-match weeks (PM), with two post-training weeks following an official match. Both matches, from the same official competition, were played at the same home venue against opponents with similar standings and had similar durations (94 vs. 96 min). As a consequence of the match calendar, ssg and ssg + GK were conducted consecutively during the first two weeks of intervention, while LSG and LSG + GK were conducted during the last two weeks, both with one NMW and PM. Two bouts were performed in the same pitch area: the first was played without (1st: ssg or LSG) and the second with GK (2nd: ssg + GK or LSG + GK). The total duration for SSGs was 3 min with a 6 min rest period between bouts. SSG formats were performed at the beginning of three different sessions with a 48 h rest between them during NMW (NMW-1, NMW-2, and NMW-3), and at 24 h (ssg/LSG-24H), 72 h (ssg/LSG-72H), and 120 h (ssg/LSG-120H) PM. Before each session, HRV, CMJ, and maximum isometric force (MIF) of the hamstrings were assessed (Figure 1). A 3 min rest period was used between the different tests. The player order used during the tests was the same for all sessions. During the jump and strength tests, players were verbally encouraged to exert their maximal effort. The physical metric averages for both session types in NMW (i.e., NMW-1, NMW-2, and NMW-3) were used as the baseline for analyzing variation throughout the PM (i.e., ssg-week and LSG-week).

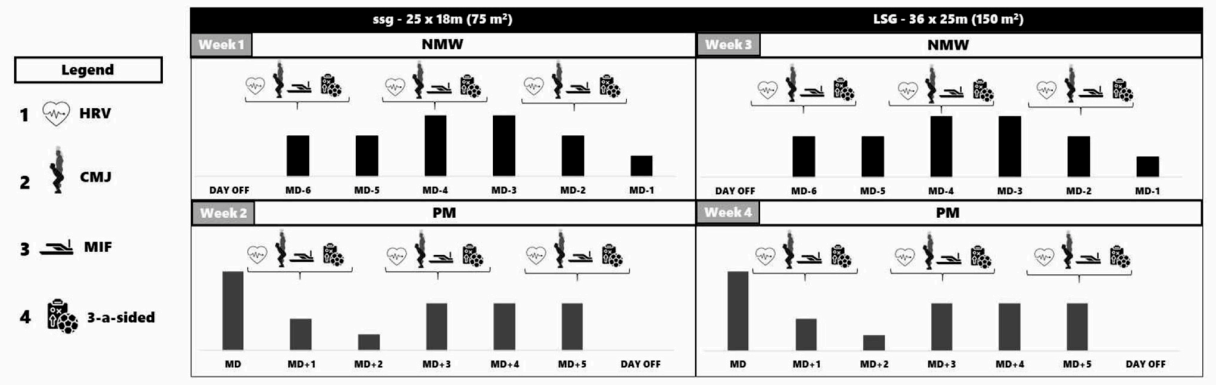


Figure 1. Schematic timeline of study methodology. NMW: non-match week (Weeks 1 and 3, featuring small (ssg) and large (LSG) formats, respectively); PM: post-match week (Weeks 2 and 4); HRV: heart rate variability test; CMJ: countermovement jump test; MIF: maximum hamstrings isometric force test; 3-a-side: 3 vs. 3-a-side game formats test. Training sessions followed a structured microcycle organization, explicitly detailing daily schedules relative to the number of days before or after a match (match day minus [MD-6 to MD-1] or match day plus [MD+1 to MD+5]) or complete rest days (DAY OFF).

2.2. Participants

Data were collected from 15 young male field soccer players (age: 18.3 ± 0.6 years; height: 178.2 ± 5.6 cm; body mass: 70.9 ± 6.4 kg) belonging to the academy of a Spanish La Liga Club. Players were assigned to sub-groups based on their average match participation across both fixtures: starters (>75 min [86.0 ± 6.9 min]; $n = 7$) and non-starters (<30 min [15.2 ± 11.4 min]; $n = 8$); with descriptive values for each group fully detailed in Table S1. Positions included were: defenders (starters: 4; non-starters: 4), midfielders (starters: 2; non-starters: 2), and forwards (starters: 1; non-starters: 2). All players were instructed to maintain their habitual routines, had been actively training for at least 3 years, and participated in five training sessions (90 min each) plus one official game per week. The procedures were fully explained to the participants and the club staff. The players (or parents of minors) gave their written consent. This study was approved by the Institutional Review Committee of Pablo de Olavide University (Ref: 25/7-82) and was performed in accordance with the Declaration of Helsinki.

2.3. Procedures

2.3.1. Reliability

The reliability of different SSG outputs and physical metrics was assessed in a convenience subsample of 18 outfield players belonging to the same U19 team across 3 separate trials. As a result of dropout ($n = 3$ players), 15 players were available for the physical metrics during the second and third tests (between-session reliability). Although players who dropped out (collected on the first occasion) were excluded from the final analysis, they were retained in the SSG formats in order to avoid altering the tactical structure. SSG outputs during NMW (i.e., NMW-1, NMW-2, and NMW-3) for each format (i.e., ssg and LSG) were included in the final sample of the within-session reliability analysis, whereas the mean values across trials within NMW sessions were analyzed for physical metrics.

2.3.2. Small-Sided Games

The SSG format consisted of 3-a-side games, with a total duration of 3 min and 6 min of rest between bouts. The first bout was played without a GK (i.e., 3 vs. 3), with the aim of free play focused on maintaining possession. In contrast, the second bout was played with GK (i.e., 3 vs. 3 + 2 GK) to defend a soccer goal (6 × 2 m) and score more goals than the opponent. The SSG format was determined using pitch dimensions, divided into small (ssg: 25 × 18 m, 75 m^2 per player) and large (LSG: 36 × 25 m, 150 m^2 per player) pitch areas. Team compositions and opponent pairings remained identical across SSG formats, having practiced these configurations weekly during the month prior to the intervention, and were composed of starter and non-starter players. The coach was required to provide verbal encouragement and immediately introduce a ball when it left the field to prevent interruptions, while no tactical substitutions were performed. Before the SSGs, players completed a standardized warm-up consisting of 4 min of soccer-specific drills using a passing drill (area: 10 × 20 m).

2.3.3. Training Load Monitoring

External load data were quantified by week type (NMW or PM) to control the training load during the SSGs (ssg and LSG). The players were monitored using a GNSS APEX (STATSports, Newry, Northern Ireland, UK) with a sampling frequency of 10 Hz. Each player wore a tight vest with the device on the back of the upper body, between the scapulae, and it was activated 15 min before the start, and the number of connected satellites ranged between 18 and 21. Doppler-derived speed data were exported from the manufacturer's software (STATSports Sonra 2.1.4, Newry, Northern Ireland, UK). The variables used to

quantify external load were selected according to usual arbitrary thresholds [20]: TD, high-speed running (HSR; meters covered faster than 19.8 $km \cdot h^{-1}$), sprinting distance (SPR; meters covered faster than 25.2 $km \cdot h^{-1}$), number of accelerations (ACC; >2 $m \cdot s^{-2}$), and decelerations (DEC; <−2 $m \cdot s^{-2}$).

2.3.4. Heart Rate Variability Measurement

To avoid circadian effects on HRV, all assessments were performed at the same time of day (between 8:00 and 8:30 a.m.), with room temperature maintained between 18 and 22 °C and dim ambient light. During the test, the subjects lay supine for at least 10 min. RMSSD was used as a parasympathetic indicator [12]. RR intervals were recorded using a V800 Polar watch paired with a Polar H7 chest strap (Polar Electro OY, Kempele, Finland). Raw recordings were carefully inspected, and ectopic beats or artifacts were identified and corrected using standard software filters, while recordings presenting excessive noise or persistent abnormalities were excluded. Subsequently, raw data were exported and analyzed using Polar FlowSync software (version 4.0.11; Polar Electro Oy, Kempele, Finland). Following a 2 min stabilization period [10], only the last 8 min of the signal were analyzed, in accordance with established protocols in soccer players [14].

2.3.5. Jumping Ability Test

Jump height was calculated to the nearest 0.1 cm from flight time measured with an infrared timing system (Optojump Next; Microgate, Bolzano, Italy). Before the assessment, players performed a standardized warm-up consisting of 5 min of jogging at a self-selected easy pace and five progressive submaximal jumps. Following previous soccer protocols [34], the subjects completed five maximal CMJs, with a 20 s rest between jumps, with their hands on their hips (without arm swing). The highest and lowest values were discarded, and the remaining values were averaged for further analysis [34].

2.3.6. Isometric Hamstrings Strength Test

The players completed a standardized warm-up that included three submaximal isometric repetitions in a glute bridge position at 30%, 40%, and 50% perceived effort, followed by two submaximal repetitions in the prone position with full extension of the knee and hip flexors on the platform. All strength tests were completed using a NordBord device (VALD Performance®, Brisbane, QLD, Australia) to determine the hamstrings MIF. Specifically, tests were performed using the ISO Prone protocol, where participants were positioned prone (lying face down on the platform) with full knee and hip extension (0° of knee flexion), and their ankles were secured immediately superior to the lateral malleolus using the device's standard ankle braces and integrated load-cell system. After the standardized warm-up, participants completed two maximal repetitions of the bilateral hamstrings prone exercise, with a 30 s rest between them. The instructions to the players were to apply maximal force from the beginning and keep it for 5 s. Bilateral (MIF) and unilateral (left: LMIF; right: RMIF) maximal hamstrings isometric forces were derived from the bilateral repetition provided by the NordBord system using VALDHub® systems software (https://hub.valdperformance.com/, VALD Performance®, Brisbane, QLD, Australia).

2.4. Statistical Analyses

Descriptive statistics are presented as the mean ± standard deviation (SD). Absolute test–retest reliability was examined using the standard error of measurement (*SEM*), expressed in relative terms through the coefficient of variation (CV). The *SEM* was calculated as the square root of the mean total within-subject variance. Relative reliability was calculated using the intraclass correlation coefficient (ICC) with a 95% confidence interval

(CI), applying a two-way random-effects model. ICC values were interpreted as follows: <0.5 = poor; 0.5–0.75 = moderate; 0.75–0.90 = good; and >0.90 = excellent [35], whereas CV was considered acceptable when <10%. To provide a practical interpretation, the smallest worthwhile change (SWC) was calculated by multiplying the between-individual SD by 0.2 [36]. The normality of the results was tested using the Shapiro–Wilk test. Linear mixed-effects models were fitted to analyze post-fatigue effects and compare responses between both weeks alongside training week variations and SSG formats. A total of 360 observations were included in each model analyzing SSG-derived variables, whereas 180 observations were included in each model analyzing HRV, CMJ, and hamstrings MIF. The balanced design indicated complete data across all measurement occasions (i.e., no missing observations). Player was included as a random intercept to account for repeated measures within individuals, assuming a variance-components covariance structure. A likelihood ratio test (LRT) compared a reduced model containing all two-way interactions against the original full factorial model (incorporating all interactions among Week, Session, Pitch Size, GK, and Player Status), with both models initially estimated by maximum likelihood (ML). Degrees of freedom and p-values for fixed effects were calculated using the Satterthwaite approximation. The final reduced model was then refitted using restricted maximum likelihood (REML) for final reporting, including 95% CIs for fixed-effect estimates, and effect sizes for each term were expressed as partial eta-squared (n_p^2). Because the two-way reduced model still tested multiple interaction terms across the five factors and across outcome variables, a Benjamini–Hochberg false discovery rate (FDR) correction was applied to the family of two-way interaction p-values (pooled across all outcome variables) to control for multiple comparisons; main effects, which corresponded to the primary prespecified hypotheses, were evaluated at the nominal alpha = 0.05 without FDR adjustment. Post-hoc pairwise comparisons for significant main effects and FDR-significant interactions were adjusted using the Bonferroni method. All analyses were performed in R (version 4.3.3) using RStudio (version 2026.07.1+147), with the packages lme4 (1.1.35.1), lmerTest (3.1.3), and emmeans (1.10.0).

3. Results

The reliability of the time-motion and fitness performance variables is presented in Table 1. Fitness variables (i.e., CMJ height, LMIF, RMIF, and MIF) showed excellent reliability values: CVs < 8% and ICCs ≥ 0.90. Among the time-motion variables, only TD showed acceptable absolute reliability (CV ~ 8%), whereas HSR and SPR exhibited extremely high variability (CVs ranging from 86% to 215% and 183% to 315%, respectively) alongside negligible or negative ICC values.

No significant differences were found for almost all weekly external loads between NMW in both groups, except a significantly lower TD during LSG-week compared to ssg-week (34,237.2 ± 2392.1 vs. 38,552.4 ± 2221.5 m, $p < 0.05$) in starters. In PM weeks, starter players covered significantly greater HSR (1257.4 ± 191.1 vs. 893.8 ± 286.9 m, $p < 0.05$) and SPR (525.1 ± 146.7 vs. 291.6 ± 160.2 m, $p < 0.05$) during LSG-week than ssg-week, while no differences were observed in non-starters (Table S2). Overall, greater values across all external load variables were observed in starters compared to non-starters during matches (Table S1). Furthermore, a match-to-match difference was exclusively found in starters, driven by an increase in SPR during the second match compared to the first (106.4 ± 70.1 vs. 155.6 ± 90.4 m, $p < 0.05$).

3.1. Time-Motion Activity During the Small-Sided Games

Time-motion activity during the SSGs is presented in Table 2. The LMM analysis revealed significant main effects for Week ($F_{1, 326} = 61.67$; $n_p^2 = 0.16$, $p < 0.001$), Session

($F_{2, 326} = 3.30$; $n_p^2 = 0.02$, $p = 0.038$), Pitch Size ($F_{1, 326} = 464.88$; $n_p^2 = 0.59$, $p < 0.001$), and GK ($F_{1, 326} = 196.32$; $n_p^2 = 0.37$, $p < 0.001$) on TD. Likewise, a Session × Pitch Size ($F_{2, 326} = 5.98$; $n_p^2 = 0.04$, $p = 0.003$) interaction was found for TD. Significant main effects were observed for Pitch Size (HSR: $F_{1, 326} = 70.86$; $n_p^2 = 0.18$, $p < 0.001$; SPR: $F_{1, 326} = 22.4$; $n_p^2 = 0.06$, $p < 0.001$) and GK (HSR: $F_{1, 326} = 33.39$; $n_p^2 = 0.09$, $p < 0.001$; SPR: $F_{1, 326} = 23.5$; $n_p^2 = 0.07$, $p < 0.001$) for high-speed metrics. ACC was significantly influenced by both Week ($F_{1, 326} = 8.33$; $n_p^2 = 0.02$, $p = 0.004$) and Pitch Size ($F_{1, 326} = 3.97$; $n_p^2 = 0.01$, $p = 0.047$). DEC exhibited a significant main effect of Week ($F_{1, 326} = 16.91$; $n_p^2 = 0.05$, $p < 0.001$). No other main effects or interaction terms reached statistical significance ($p > 0.05$).

Table 1. Test–retest reliability of external load in 3-a-side SSG format and physical performance.

Variable	Format	ICC	CI (95%)	*SEM*	CV	SWC (95% CI)
TD	ssg	0.58	0.01–0.85	23.4	6.4%	4.1 (3.1–5.4)
	ssg + GK	0.50	−0.19–0.82	23.6	7.3%	4.2 (3.1–5.4)
	LSG	0.59	0.03–0.85	29.1	6.9%	4.8 (3.2–6.6)
	LSG + GK	0.55	−0.07–0.84	24.9	6.6%	4.6 (3.6–5.6)
HSR	ssg	−0.02	−1.41–0.63	1.7	215.0%	0.2 (0.1–0.4)
	ssg + GK	0.58	−0.01–0.85	2.0	139.1%	0.4 (0.2–0.6)
	LSG	0.49	−0.22–0.81	3.6	125.3%	0.6 (0.3–0.9)
	LSG + GK	0.11	−1.12–0.68	7.6	86.4%	1.8 (1.4–2.3)
SPR	ssg	-	-	-	-	0.0 (0.0–0.7)
	ssg + GK	−0.42	−2.10–0.65	1.6	314.9%	0.1 (0.0–0.2)
	LSG	−0.31	−2.12–0.52	0.2	239.0%	0.0 (0.0–0.1)
	LSG + GK	−0.71	−3.01–0.38	3.9	183.3%	0.6 (0.4–0.9)
ACC	ssg	0.77	0.46–0.92	3.1	20.4%	0.6 (0.4–0.7)
	ssg + GK	−0.33	−2.15–0.52	3.8	25.0%	0.7 (0.6–0.8)
	LSG	0.27	−0.74–0.73	4.2	29.1%	0.7 (0.4–1.0)
	LSG + GK	0.48	−0.24–0.81	2.7	18.7%	0.5 (0.4–0.6)
DEC	ssg	0.59	0.01–0.85	3.5	20.6%	0.6 (0.5–0.8)
	ssg + GK	0.26	−0.77–0.73	3.7	25.7%	0.7 (0.4–0.7)
	LSG	0.58	−0.01–0.85	3.0	20.0%	0.5 (0.4–0.7)
	LSG + GK	0.61	0.07–0.86	3.1	21.4%	0.6 (0.4–0.7)
CMJ		0.99	0.97–0.99	0.4	1.3%	0.5 (0.3–0.7)
LMIF		0.93	0.84–0.98	23.9	7.1%	11.1 (7.1–15.1)
RMIF		0.94	0.86–0.98	24.5	7.8%	8.7 (5.6–11.8)
MIF		0.96	0.90–0.98	44.5	7.1%	15.2 (9.7–20.6)

ICC: intraclass correlation coefficient; CI: confidence interval (95%); *SEM*: standard error of measurement; CV: coefficient of variation; SWC: smallest worthwhile change. TD: total distance; HSR: high-speed running; SPR: sprint distance; ACC: accelerations; DEC: decelerations. ssg: 3-a-side game without a goalkeeper (GK) in a relative player area of 75 m^2; ssg + GK: 3-a-side game with GK in a relative player area of 75 m^2; LSG: 3-a-side game without GK in a relative player area of 150 m^2; LSG + GK: 3-a-side game with GK in a relative player area of 150 m^2; CMJ: countermovement jump height; LMIF: left-leg maximum isometric force; RMIF: right-leg maximum isometric force; and MIF: bilateral maximum isometric force.

Post-hoc pairwise comparisons showed that overall running volume was significantly lower during the PM than during NMW for TD ($\beta = -21.75$ m, *SEM* = 2.77, 95% CI [−27.16, −16.35]; $p < 0.001$), ACC ($\beta = -1.02$ counts, *SEM* = 0.35, 95% CI [−1.70, −0.32]; $p = 0.004$), and DEC ($\beta = -1.46$ counts, *SEM* = 0.35, 95% CI [−2.15, −0.76]; $p < 0.001$). TD was significantly higher at 72H-PM than at 120H-PM in ssg-week ($\beta = 15.99$ m, *SEM* = 4.79, 95% CI [6.59, 25.38]; $p = 0.014$).

3.2. Heart Rate Variability and Physical Performance

The LMM analysis for HRV (Figure 2) revealed no significant main effects for Week or Session. However, a significant interaction for Session × Player Status ($F_{2, 326} = 11.91$;

$n_p^2 = 0.07$, $p < 0.001$) was identified. Post-hoc analyses indicated that starters exhibited an HRV fluctuation between the first and second session (β = 12.25 ms, *SEM* = 3.30, 95% CI [5.78,18.72]; $p = 0.004$), which was conversely observed in non-starters (β = −9.65 ms, *SEM* = 3.01, 95% CI [−15.70, −3.60]; $p = 0.028$).

Table 2. Evolution of time-motion activity across small-sided game (SSG) formats over different training weeks.

Variable	Format	Group	NMW-1	NMW-2	NMW-3	24H-PM	72H-PM	120H-PM
TD (m) [a,b,c,d,¶,*]	ssg	Starters	366.5 ± 40.6	380.6 ± 16.2	378.4 ± 19.8	335.5 ± 15.7	350.9 ± 35.6	328.2 ± 9.3 [¥]
		Non-Starters	355.3 ± 16.0	360.7 ± 25.7	349.6 ± 36.6	331.2 ± 36.5	350.3 ± 27.4	296.8 ± 28.5 [¥]
	ssg + GK	Starters	315.0 ± 25.5	341.8 ± 23.1	314.1 ± 25.4	310.9 ± 25.0	304.9 ± 28.0	311.5 ± 33.6 [¥]
		Non-Starters	320.8 ± 22.2	332.5 ± 20.9	316.1 ± 31.9	301.7 ± 25.8	303.2 ± 21.5	300.9 ± 43.5 [¥]
	LSG	Starters	428.1 ± 18.3	415.9 ± 44.9	441.3 ± 39.3	431.1 ± 33.9	401.0 ± 44.2	400.0 ± 27.1
		Non-Starters	425.9 ± 32.8	408.0 ± 32.7	408.6 ± 36.9	413.7 ± 32.6	393.2 ± 22.1	389.1 ± 20.0
	LSG + GK	Starters	387.1 ± 19.2	376.4 ± 23.7	397.6 ± 37.7	358.1 ± 26.9	369.9 ± 35.3	368.0 ± 34.4
		Non-Starters	385.9 ± 29.8	372.0 ± 23.6	357.1 ± 29.1	357.9 ± 18.9	360.1 ± 30.1	345.0 ± 24.7
HSR (m) [c,d]	ssg	Starters	0.2 ± 0.3	2.3 ± 3.6	1.2 ± 2.5	0.0 ± 0.0	1.3 ± 3.5	0.0 ± 0.0
		Non-Starters	0.2 ± 0.3	1.1 ± 1.8	0.5 ± 1.1	0.0 ± 0.0	0.5 ± 0.7	3.3 ± 6.6
	ssg + GK	Starters	0.7 ± 1.3	1.6 ± 2.9	3.9 ± 6.4	2.3 ± 2.1	3.1 ± 3.7	0.3 ± 0.5
		Non-Starters	0.1 ± 0.2	2.8 ± 3.0	1.7 ± 2.6	2.8 ± 4.0	1.9 ± 3.9	2.5 ± 4.1
	LSG	Starters	0.0 ± 0.1	2.1 ± 4.8	1.7 ± 4.2	0.0 ± 0.0	3.1 ± 7.0	1.4 ± 1.9
		Non-Starters	0.1 ± 0.2	4.1 ± 3.6	4.2 ± 7.3	1.1 ± 2.1	5.2 ± 6.4	9.2 ± 9.3
	LSG + GK	Starters	11.1 ± 5.7	13.7 ± 11.8	14.1 ± 13.5	8.1 ± 9.1	4.1 ± 7.7	8.0 ± 8.4
		Non-Starters	13.9 ± 14.8	5.3 ± 4.2	8.0 ± 7.4	7.6 ± 9.5	6.7 ± 6.1	13.3 ± 10.5
SPR (m) [c,d]	ssg	Starters	0.0 ± 0.0	0.4 ± 1.0	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0
		Non-Starters	0.0 ± 0.0	0.1 ± 0.2	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0
	ssg + GK	Starters	0.0 ± 0.0	0.0 ± 0.0	1.7 ± 2.9	0.1 ± 0.2	0.1 ± 0.3	0.0 ± 0.0
		Non-Starters	0.0 ± 0.0	0.1 ± 0.2	0.4 ± 1.0	0.2 ± 0.5	0.3 ± 1.0	0.2 ± 0.5
	LSG	Starters	1.9 ± 2.6	0.1 ± 0.3	0.1 ± 0.3	1.0 ± 2.2	0.2 ± 0.5	0.0 ± 0.0
		Non-Starters	3.5 ± 3.6	0.0 ± 0.0	0.2 ± 0.5	5.3 ± 5.0	0.7 ± 1.7	0.3 ± 0.5
	LSG + GK	Starters	1.7 ± 1.7	2.1 ± 3.4	4.7 ± 6.1	1.5 ± 2.3	2.1 ± 5.6	1.4 ± 3.1
		Non-Starters	2.2 ± 4.6	1.0 ± 2.2	1.3 ± 1.9	1.7 ± 3.0	1.0 ± 1.5	2.3 ± 3.7
ACC (*n*) [a,c,*]	ssg	Starters	15.4 ± 4.2	13.6 ± 3.4	15.0 ± 4.2	12.9 ± 4.1	14.0 ± 2.2	12.2 ± 4.5
		Non-Starters	15.0 ± 4.1	14.3 ± 5.3	16.3 ± 5.5	15.7 ± 4.1	14.9 ± 5.8	10.6 ± 6.0
	ssg + GK	Starters	14.3 ± 2.7	15.8 ± 3.0	15.6 ± 4.2	15.2 ± 1.9	14.4 ± 2.8	14.6 ± 2.5
		Non-Starters	14.3 ± 3.6	15.3 ± 4.3	15.0 ± 4.4	15.7 ± 2.6	15.3 ± 2.4	13.9 ± 3.3
	LSG	Starters	17.4 ± 5.9	14.0 ± 2.4	14.0 ± 2.2	12.7 ± 3.6	12.6 ± 5.0	11.6 ± 4.3
		Non-Starters	15.1 ± 4.1	14.5 ± 3.9	12.5 ± 6.2	12.4 ± 4.1	14.9 ± 2.7	13.9 ± 4.2
	LSG+GK	Starters	14.6 ± 4.0	14.7 ± 2.8	15.7 ± 2.3	13.1 ± 2.1	14.1 ± 1.8	13.1 ± 3.0
		Non-Starters	13.5 ± 3.4	14.0 ± 2.7	13.0 ± 3.2	12.9 ± 2.6	14.7 ± 2.9	13.3 ± 2.8
DEC (*n*) [a,*]	ssg	Starters	17.3 ± 2.3	16.3 ± 4.1	18.5 ± 3.1	11.9 ± 2.3	14.0 ± 3.1	13.0 ± 2.6
		Non-Starters	15.9 ± 3.4	17.0 ± 6.8	17.5 ± 4.8	14.0 ± 3.4	15.5 ± 4.3	12.8 ± 4.7
	ssg + GK	Starters	12.9 ± 2.4	15.1 ± 4.6	14.9 ± 3.7	15.1 ± 2.3	15.0 ± 2.9	14.2 ± 1.8
		Non-Starters	16.3 ± 2.9	12.5 ± 5.3	15.1 ± 3.8	13.5 ± 2.5	15.4 ± 3.2	12.4 ± 3.5
	LSG	Starters	17.4 ± 2.1	14.7 ± 2.8	15.4 ± 2.8	14.7 ± 2.6	12.7 ± 4.0	13.0 ± 3.1
		Non-Starters	15.9 ± 2.5	13.5 ± 3.4	13.1 ± 5.1	13.3 ± 5.7	15.4 ± 2.9	13.6 ± 5.1
	LSG + GK	Starters	13.7 ± 1.3	15.4 ± 2.6	16.1 ± 4.5	12.4 ± 1.4	14.7 ± 4.3	13.6 ± 2.8
		Non-Starters	14.0 ± 3.2	15.0 ± 4.8	13.1 ± 5.2	12.8 ± 2.0	15.1 ± 3.8	13.5 ± 3.5

NMW-1: First test of non-match week (NMW); NMW-2: Second test of NMW; NMW-3: Third test of NMW; 24H-PM: First test 24 h post-match (PM); 72H-PM: Second test 72 h PM; 120H-PM: Third test 120 h PM. TD: total distance covered; HSR: high-speed running; SPR: sprint distance; ACC: accelerations; DEC: decelerations. ssg: 3-a-side game without GK in a relative player area of 75 m^2; ssg + GK: 3-a-side game with GK in a relative player area of 75 m^2; LSG: 3-a-side game without GK in a relative player area of 150 m^2; LSG + GK: 3-a-side game with GK in a relative player area of 150 m^2. Fixed effects: [a] Week; [b] Session; [c] Pitch Size; [d] GK. [¶] Session × Pitch Size ($p < 0.05$). Post-hoc: * Week (NMW vs. PM); [¥] Sessions (72H-PM vs. 120H-PM).

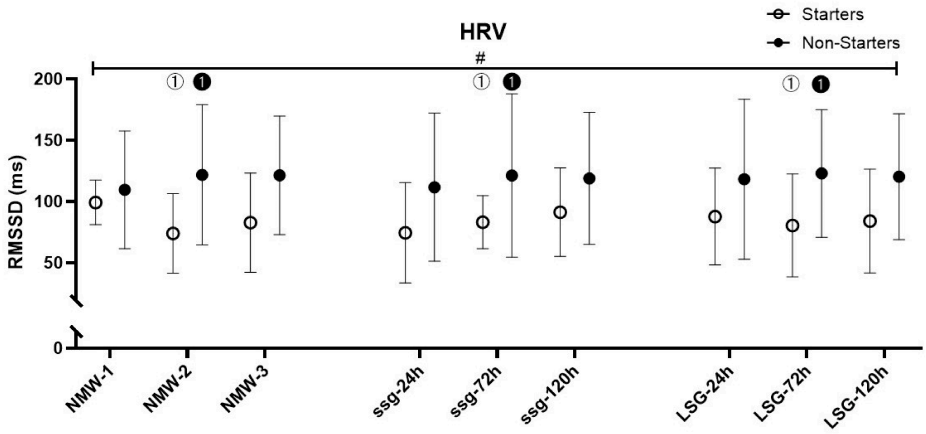


Figure 2. Changes in root mean square of successive differences (RMSSD) through a training week (NMW: non-match week) and during post-match weeks using different small-sided games formats (ssg; 25 × 18 m pitch, 75 m^2 vs. LSG; 36 × 25 m pitch, 150 m^2) at 24, 72, and 120 h post-match. Fixed effects: # Session × Player Status ($p < 0.001$). Post-hoc: ①, ❶ Intra-group (compared to first session in starters and non-starters; $p < 0.05$).

CMJ analysis (Figure 3) revealed significant main fixed effects for Week ($F_{1, 326} = 38.42$; $n_p^2 = 0.11$, $p < 0.001$) and Session ($F_{2, 326} = 8.26$; $n_p^2 = 0.05$, $p < 0.001$). Subsequent post-hoc analyses indicated that CMJ height was significantly greater during NMW than during PM (β = 0.90 cm, *SEM* = 0.15, 95% CI [0.61, 1.18]; $p < 0.001$). Specifically, higher CMJ height was observed during LSG-week in NMW compared to PM (β = 1.42 cm, *SEM* = 0.21, 95% CI [1.01, 1.81]; $p < 0.001$). Lower CMJ height values were found in the first session compared to the second (β = −0.65 cm, *SEM* = 0.18, 95% CI [−1.01, −0.31]; $p < 0.001$) and third session (β = −0.59 cm, *SEM* = 0.18, 95% CI [−0.94, −0.24]; $p < 0.001$).

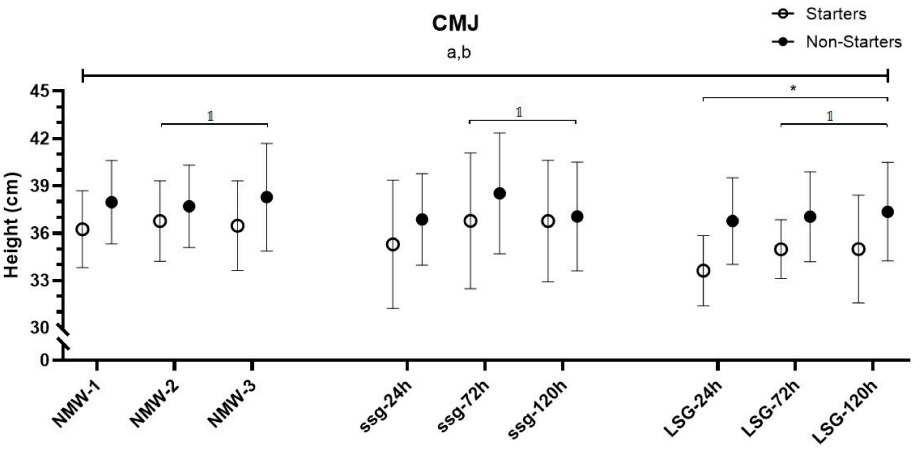


Figure 3. Changes in countermovement jump (CMJ) height (cm) through a training week (NMW: non-match week) and during post-match weeks using different small-sided games formats (ssg; 25 × 18 m pitch [75 m^2] vs. LSG; 36 × 25 m pitch [150 m^2]) at 24, 72, and 120 h post-match. Fixed effects: [a] Week; [b] Session ($p < 0.001$). Post-hoc: * Week (NMW vs. PM; $p < 0.001$). 𝟙 Sessions (compared to first session; $p < 0.001$).

Regarding hamstrings MIF metrics (Figure 4), the LMM analysis showed significant main fixed effects for Week (LMIF: $F_{1, 326} = 14.84$; $n_p^2 = 0.04$, $p < 0.001$; RMIF: $F_{1, 326} = 4.35$; $n_p^2 = 0.01$, $p = 0.038$; MIF: $F_{1, 326} = 11.68$; $n_p^2 = 0.03$, $p < 0.001$), Session (RMIF: $F_{2, 326} = 7.40$; $n_p^2 = 0.04$, $p < 0.001$; MIF: $F_{2, 326} = 3.80$; $n_p^2 = 0.02$, $p = 0.023$), and Pitch Size (LMIF: $F_{1, 326} = 12.41$; $n_p^2 = 0.04$, $p < 0.001$; RMIF: $F_{1, 326} = 8.75$; $n_p^2 = 0.03$, $p = 0.003$; MIF: $F_{1, 326} = 13.61$; $n_p^2 = 0.04$, $p < 0.001$). Significant Week × Pitch Size (LMIF: $F_{1, 326} = 12.01$;

$n_p^2 = 0.04$, $p = 0.010$; RMIF: $F_{1, 326} = 8.43$; $n_p^2 = 0.03$, $p = 0.036$; MIF: $F_{1, 326} = 13.14$; $n_p^2 = 0.04$, $p = 0.009$), Session × Pitch Size (RMIF: $F_{2, 326} = 6.11$; $n_p^2 = 0.04$, $p = 0.027$; MIF: $F_{2, 326} = 6.29$; $n_p^2 = 0.04$, $p = 0.025$) and Session × Player Status (RMIF: $F_{2, 326} = 7.70$; $n_p^2 = 0.05$, $p = 0.010$) interactions were observed.

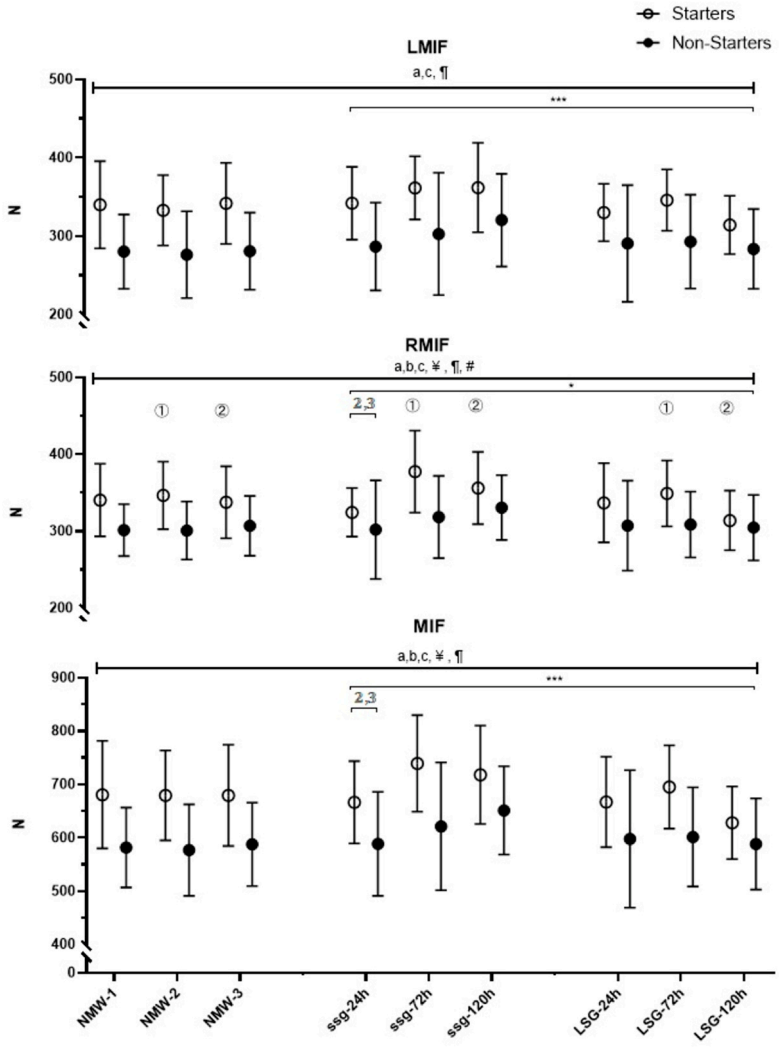


Figure 4. Changes in maximum isometric force through a training week (NMW: non-match week) and during post-match weeks using different small-sided games formats (ssg; 25 × 18 m pitch, 75 m^2 vs. LSG; 36 × 25 m pitch, 150 m^2) at 24, 72, and 120 h post-match. LMIF: left-leg maximum isometric force; RMIF: right-leg maximum isometric force; and MIF: bilateral maximum isometric force. Fixed effects: [a] Week; [b] Session; [c] Pitch Size. [¥] Week × Pitch Size ($p < 0.05$). [¶] Session × Pitch Size ($p < 0.05$). [#] Session × Player Status ($p < 0.05$). Post-hoc: *, *** Week (NMW vs. PM; $p < 0.05$, 0.001); [②,③] Sessions (compared to second and third session; $p < 0.01$); ①, ② Intra-group (compared to first and second session in starters; $p < 0.01$).

Post-hoc analyses confirmed that force outputs were consistently inferior in NMW compared to PM (LMIF: β = −10.72 N, *SEM* = 2.78, 95% CI [−16.18, −5.26], $p < 0.001$; RMIF: β = −5.22 N, *SEM* = 2.51, 95% CI [−10.13, −0.31], $p = 0.038$; MIF: β = −15.95 N, *SEM* = 4.67, 95% CI [−25.10, −6.80]; $p < 0.001$), as well as in ssg-week for first session compared to second (RMIF: β = −18.55 N, *SEM* = 4.33, 95% CI [−27.05, −10.06]; $p < 0.001$; MIF: β = −24.73 N, *SEM* = 8.07, 95% CI [−40.56, −8.91]; $p = 0.036$) and third session (RMIF: β = −15.58 N, *SEM* = 4.33, 95% CI [−24.08, −7.01]; $p = 0.006$; MIF: β = −29.58 N, *SEM* = 8.07, 95% CI [−45.40, −13.75]; $p = 0.004$). Significant intra-group variability was also detected. Specifically, starters exhibited a significantly lower RMIF in the first compared

with the second session (β = −19.36 N, *SEM* = 4.48, 95% CI [−28.14, −10.58]; $p < 0.001$), and higher force output in second compared to third session (β = 18.63 N, *SEM* = 4.48, 95% CI [9.84, 27.41]; $p < 0.001$), whereas no variations were observed in non-starters.

4. Discussion

To our knowledge, this is the first study to examine whether different SSG formats are sensitive to post-match fatigue by analyzing changes in time-motion performance and physiological and athletic markers across a typical in-season training week in elite youth soccer players. The main finding was that although post-match fatigue was associated with reductions in several external-load variables during SSGs, particularly TD, ACC, and DEC, the practical value of these changes was substantially limited by the poor reliability and high variability of many SSG-derived metrics. In contrast, neuromuscular assessment consistently detected post-match fatigue, as reflected by reductions in CMJ height, whereas hamstrings MIF responses varied by player status. Starters showed progressively higher force values as the week progressed, a pattern not observed in non-starters. Meanwhile, HRV showed divergent responses between groups, with RMSSD decreasing in starters but increasing in non-starters during the middle of the training week. Collectively, these findings suggest that, despite some responsiveness to post-match fatigue, short 3-a-side SSGs have limited utility as standalone monitoring tools and should complement rather than replace direct neuromuscular assessments when evaluating recovery status in elite youth soccer players.

The use of SSGs for assessing physical performance has been debated because, unlike standardized physical tests, performance during SSGs emerges from the interaction of physical, technical, tactical, and contextual factors [15–18]. Therefore, the usefulness of SSGs for monitoring fatigue depends not only on their responsiveness to physiological changes but also on the reliability of the measured variables. In the present study, TD showed acceptable absolute reliability (CV: ~8%) but only moderate relative reliability (ICC: ~0.53) in 3-a-side SSGs played with or without GK across different pitch sizes, consistent with the findings of Riboli et al. [18], who likewise concluded that SSG-derived TD was insufficiently reliable to monitor changes in aerobic fitness. Despite these limitations, a consistent reduction in TD covered during post-match weeks was observed across most SSG formats, supporting the notion that match play induces residual fatigue that influences subsequent training performance. In addition to TD, other mechanical load indicators were also affected, as reflected by reduced ACC and DEC during PM weeks compared with NMW. These findings align with previous research showing decrements in running performance during congested or post-match periods [37,38]. However, the magnitude and timing of these reductions varied considerably depending on pitch size, GK inclusion, and player status, suggesting that the sensitivity of SSG-derived metrics is highly dependent on task constraints. In this regard, SSG outputs exhibited substantial inconsistency across sessions. This variability likely reflects the inherent stochasticity of SSGs, in which technical-tactical behaviors, player interactions, and contextual constraints strongly influence physical outputs [25], limiting their use as a standalone tool for monitoring fatigue. However, technical or tactical behaviors within each SSG format were not quantified. Thus, without tracking these contextual and behavioral dynamics, it is difficult to determine whether lower SSG outputs reflect true post-match fatigue or simply alterations in game behavior and tactical pacing. In particular, the greatest limitation was evident in distances covered at high speed and sprinting. These variables exhibited poor relative reliability and very large coefficients of variation, likely reflecting the restricted opportunities to reach high running speeds during 3 vs. 3 SSGs played on relatively small pitches. Given this substantial measurement noise, inferential testing of minor variations in high-intensity parameters lacks

practical meaning. Consequently, the observed floor effects reduce both the reproducibility and practical utility of these metrics for monitoring post-match fatigue in this specific training context. Although these variables were retained because they represent standard GNSS-derived external-load metrics routinely quantified in soccer and allow comparisons with previous literature, the excessive measurement noise severely limits their practical utility as fatigue-monitoring tools.

In contrast to the SSG-derived metrics, CMJ height proved to be a sensitive and reliable indicator of post-match neuromuscular fatigue. With an excellent reliability profile, the typical measurement error was substantially lower than the observed peak decrement, ensuring that changes reflect true neuromuscular alterations rather than technical noise. A significant reduction in CMJ height was observed during post-match weeks compared with non-match weeks. These findings are in line with previous studies reporting impairments in jump performance following competitive soccer matches [1,39], reinforcing the value of CMJ as a simple and practical marker of neuromuscular recovery. Interestingly, the largest impairments were accentuated during LSG-week, coinciding with higher high-speed running demands during the match, highlighting the potential sensitivity of CMJ height to variations in match load. Alternatively, worse CMJ height was observed at the beginning of training weeks, followed by progressive improvements across subsequent sessions, suggesting that the first 72 h of the week may represent a period of acute neuromuscular recovery. Furthermore, no differences were observed according to player status, which may be partly attributable to the relatively small sample size. Therefore, these findings should be interpreted cautiously, particularly given the limited number of participants and matches examined. Nonetheless, the greater sensitivity of CMJ compared with SSG-derived variables is likely explained by its highly standardized nature. Unlike performance during SSGs, which is influenced by technical, tactical, and contextual factors, CMJ may primarily reflect changes in neuromuscular function with minimal influence from external constraints, provided that technical execution remains unchanged. Consequently, day-to-day variability is substantially lower, allowing fatigue-related changes to be detected more consistently. Collectively, these findings support CMJ as one of the most practical and robust tools for monitoring acute neuromuscular fatigue in male academy soccer players.

Hamstrings MIF showed significant post-match changes, but they did not follow the expected fatigue-related pattern, with force generally higher during PM than NMW. These improvements in hamstrings force were consistently observed across post-match sessions. Starters exhibited a progressive increase in RMIF during the first part of the week, reaching peak values at the mid-week session, followed by a slight reduction in the final session. These findings contrast with previous studies reporting reductions in hamstrings force after matches [6,8]. Nonetheless, this mid-week improvement in hamstrings force observed in starters contrasts with the absence of within-week variations in non-starters, reinforcing the importance of individualizing post-match strategies according to playing time [1,40]. Interestingly, hamstrings MIF were generally higher in post-match weeks compared to the non-match weeks. However, this effect may have been inflated by several factors, such as increased familiarization with the tests or MIF potentiation by match-play [41]. Another possible explanation is greater exposure to high-speed running during match play, which may affect the strength of the posterior thigh muscles, given their role during sprinting [42]. Collectively, these findings highlight the importance of integrating neuromuscular testing within training monitoring to better understand post-match changes in movement capacity.

HRV showed limited sensitivity to post-match fatigue in the present study. Given the lack of general main effects and the observed variation during non-match weeks, changes in RMSSD should be interpreted with caution. A significant decrease was observed in starters at mid-week, while opposite responses were detected in non-starters. It is possible that

the different match and training exposure among players increased the detectability of autonomic disturbances. This result contrasts with Muñoz et al. [14], who reported that parasympathetic function decreased 24 h after a soccer match in starter players but was fully restored at 72 h. This result also agrees with studies reporting changes in parasympathetic activity during post-match weeks [43,44] but contrasts with others showing limited sensitivity of HRV to short-term fatigue in young or well-trained populations [11]. Although autonomic responses appeared transient, the substantial natural variability observed across conditions warrants a conservative interpretation of these findings. Therefore, HRV should not be used in isolation to infer neuromuscular recovery status; rather, it should be integrated with performance-based measures, particularly when monitoring recovery in elite youth soccer players.

4.1. Study Limitations

The current study had several limitations that should be acknowledged. First, the relatively small sample size, combined with the complexity of the repeated-measures model and the number of dependent variables analyzed, may have increased the risk of both type I and type II errors. Therefore, higher-order interactions should be interpreted with caution and treated as exploratory. Second, the different SSG formats were not randomized or counterbalanced across the study period due to the constraints imposed by the competitive match calendar. Consequently, the observed differences between SSG formats cannot be attributed exclusively to pitch size or GK inclusion, as they may also have been influenced by temporal factors such as seasonal progression or accumulated fatigue. Nevertheless, weekly external training load was continuously monitored using GNSS technology, allowing us to verify that the overall training load was largely comparable between weeks. Additionally, accelerometry-derived variables were not analyzed during SSGs, which could have provided more insight into changes in movement strategies. Future research should investigate whether combining SSG performance with accelerometry metrics or personalized benchmarks enhances fatigue detection, and consider counterbalanced format sequencing with longer and repeated SSG sessions as well as format sequencing (e.g., order of GK presence) to improve sensitivity.

4.2. Practical Applications

While SSGs remain a valuable training tool, the high variability of the time-motion parameters limits their use as standalone fatigue indicators for routine monitoring. Conversely, CMJ height is a sensitive marker of post-match neuromuscular fatigue in male academy soccer players, whereas hamstrings MIF may provide complementary information on post-match responses. Given the exploratory nature of this four-week study, integrating these reliable performance-based tests with external load monitoring may provide a more robust approach for evaluating recovery and informing individualized training strategies in elite youth soccer players, while broader clinical applications should be interpreted with caution.

5. Conclusions

In conclusion, post-match changes in male academy soccer players were most consistently reflected in CMJ height, which demonstrated high reliability and decreased during post-match weeks. Hamstrings MIF also demonstrated excellent reliability, although its post-match responses were less straightforward and did not consistently reflect the expected fatigue-related pattern. In contrast, HRV showed divergent mid-week responses between starters and non-starters and did not consistently reflect neuromuscular recovery status. Although time-motion performance during 3-a-side SSGs was generally reduced during

post-match weeks, the high variability and poor reproducibility of several time-motion variables, particularly high-speed running and sprint distances, limit their usefulness as indicators of post-match fatigue in short 3-a-side SSGs formats. Consequently, SSGs may complement fatigue testing, but should not replace direct neuromuscular assessments when monitoring post-match recovery in elite male academy soccer players.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/sports14110487/s1, Table S1: Descriptive statistics of match load; Table S2: Descriptive statistics of training week load.

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Informed Consent Statement: Written informed consent was obtained from the participants (or parents of minors).

Data Availability Statement: The raw data supporting the conclusions of this article will be made available by the authors on request.

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Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Silva, J.R.; Rumpf, M.C.; Hertzog, M.; Castagna, C.; Farooq, A.; Girard, O.; Hader, K. Acute and Residual Soccer Match-Related Fatigue: A Systematic Review and Meta-analysis. *Sports Med*. **2018**, *48*, 539–583. [CrossRef] [PubMed]
2. Impellizzeri, F.M.; Marcora, S.M.; Coutts, A.J. Internal and External Training Load: 15 Years On. *Int. J. Sports Physiol. Perform*. **2019**, *14*, 270–273. [CrossRef] [PubMed]
3. Kalkhoven, J.T.; Watsford, M.L.; Coutts, A.J.; Edwards, W.B.; Impellizzeri, F.M. Training Load and Injury: Causal Pathways and Future Directions. *Sports Med*. **2021**, *51*, 1137–1150. [CrossRef] [PubMed]
4. Bourdon, P.C.; Cardinale, M.; Murray, A.; Gastin, P.; Kellmann, M.; Varley, M.C.; Gabbett, T.J.; Coutts, A.J.; Burgess, D.J.; Gregson, W.; et al. Monitoring Athlete Training Loads: Consensus Statement. *Int. J. Sports Physiol. Perform*. **2017**, *12*, S2161–S2170. [CrossRef] [PubMed]
5. Halson, S.L. Monitoring training load to understand fatigue in athletes. *Sports Med*. **2014**, *44*, 139–147. [CrossRef] [PubMed]
6. McCall, A.; Nedelec, M.; Carling, C.; Le Gall, F.; Berthoin, S.; Dupont, G. Reliability and sensitivity of a simple isometric posterior lower limb muscle test in professional football players. *J. Sports Sci*. **2015**, *33*, 1298–1304. [CrossRef] [PubMed]
7. Claudino, J.G.; Cronin, J.; Mezencio, B.; McMaster, D.T.; McGuigan, M.; Tricoli, V.; Amadio, A.C.; Serrao, J.C. The countermovement jump to monitor neuromuscular status: A meta-analysis. *J. Sci. Med. Sport* **2017**, *20*, 397–402. [CrossRef] [PubMed]
8. Franceschi, A.; Robinson, M.A.; Owens, D.; Brownlee, T.; Ferrari Bravo, D.; Enright, K. Reliability and sensitivity to change of post-match physical performance measures in elite youth soccer players. *Front. Sports Act. Living* **2023**, *5*, 1173621. [CrossRef] [PubMed]
9. Evans, D.A.; Jackson, D.T.; Kelly, A.L.; Williams, C.A.; McAuley, A.B.; Knapman, H.; Morgan, P.T. Monitoring postmatch fatigue during a competitive season in elite youth soccer players. *J. Athl. Train*. **2022**, *57*, 184–190. [CrossRef] [PubMed]
10. Flatt, A.A.; Howells, D. Effects of varying training load on heart rate variability and running performance among an Olympic rugby sevens team. *J. Sci. Med. Sport* **2019**, *22*, 222–226. [CrossRef] [PubMed]
11. Thorpe, R.T.; Strudwick, A.J.; Buchheit, M.; Atkinson, G.; Drust, B.; Gregson, W. Monitoring Fatigue During the In-Season Competitive Phase in Elite Soccer Players. *Int. J. Sports Physiol. Perform*. **2015**, *10*, 958–964. [CrossRef] [PubMed]
12. Mirto, M.; Filipas, L.; Altini, M.; Codella, R.; Meloni, A. Heart Rate Variability in Professional and Semiprofessional Soccer: A Scoping Review. *Scand. J. Med. Sci. Sports* **2024**, *34*, e14673. [CrossRef] [PubMed]

13. Sanchez-Sanchez, J.; Botella, J.; Felipe Hernandez, J.L.; Leon, M.; Paredes-Hernandez, V.; Colino, E.; Gallardo, L.; Garcia-Unanue, J. Heart Rate Variability and Physical Demands of In-Season Youth Elite Soccer Players. *Int. J. Environ. Res. Public Health* **2021**, *18*, 1391. [CrossRef] [PubMed]
14. Munoz-Lopez, A.; Nakamura, F.; Naranjo Orellana, J. Soccer Matches but Not Training Sessions Disturb Cardiac-Autonomic Regulation During National Soccer Team Training Camps. *Res. Q. Exerc. Sport* **2021**, *92*, 43–51. [CrossRef] [PubMed]
15. Owen, A.L.; Newton, M.; Shovlin, A.; Malone, S. The Use of Small-Sided Games as an Aerobic Fitness Assessment Supplement Within Elite Level Professional Soccer. *J. Hum. Kinet.* **2020**, *71*, 243–253. [CrossRef] [PubMed]
16. Milanovic, Z.; Rada, A.; Erceg, M.; Trajkovic, N.; Stojanovic, E.; Lesnik, B.; Krustrup, P.; Randers, M.B. Reproducibility of Internal and External Training Load During Recreational Small-Sided Football Games. *Res. Q. Exerc. Sport* **2020**, *91*, 676–681. [CrossRef] [PubMed]
17. Stevens, T.G.; De Ruiter, C.J.; Beek, P.J.; Savelsbergh, G.J. Validity and reliability of 6-a-side small-sided game locomotor performance in assessing physical fitness in football players. *J. Sports Sci.* **2016**, *34*, 527–534. [CrossRef] [PubMed]
18. Riboli, A.; Dellal, A.; Esposito, F.; Coratella, G. Can small-sided games assess the training-induced aerobic adaptations in elite football players? *J. Sports Med. Phys. Fit.* **2022**, *62*, 1313–1322. [CrossRef] [PubMed]
19. Younesi, S.; Rabbani, A.; Clemente, F.M.; Silva, R.; Sarmento, H.; Figueiredo, A.J. Relationships Between Aerobic Performance, Hemoglobin Levels, and Training Load During Small-Sided Games: A Study in Professional Soccer Players. *Front. Physiol.* **2021**, *12*, 649870. [CrossRef] [PubMed]
20. Beato, M.; Wren, C.; de Keijzer, K.L. The Interunit Reliability of Global Navigation Satellite Systems Apex (STATSports) Metrics During a Standardized Intermittent Running Activity. *J. Strength Cond. Res.* **2024**, *38*, e49–e55. [CrossRef] [PubMed]
21. Rowell, A.E.; Aughey, R.J.; Clubb, J.; Cormack, S.J. A Standardized Small Sided Game Can Be Used to Monitor Neuromuscular Fatigue in Professional A-League Football Players. *Front. Physiol.* **2018**, *9*, 1011. [CrossRef] [PubMed]
22. Kelly, S.J.; Murphy, A.J.; Watsford, M.L.; Austin, D.; Rennie, M. Reliability and validity of sports accelerometers during static and dynamic testing. *Int. J. Sports Physiol. Perform.* **2015**, *10*, 106–111. [CrossRef] [PubMed]
23. Shushan, T.; McLaren, S.J.; Buchheit, M.; Scott, T.J.; Barrett, S.; Lovell, R. Submaximal Fitness Tests in Team Sports: A Theoretical Framework for Evaluating Physiological State. *Sports Med.* **2022**, *52*, 2605–2626. [CrossRef] [PubMed]
24. Asian-Clemente, J.; Rabano-Muñoz, A.; Muñoz, B.; Franco, J.; Suarez-Arrones, L. Can small-side games provide adequate high-speed training in professional soccer? *Int. J. Sports Med.* **2021**, *42*, 523–528. [CrossRef] [PubMed]
25. Hill-Haas, S.V.; Dawson, B.; Impellizzeri, F.M.; Coutts, A.J. Physiology of small-sided games training in football: A systematic review. *Sports Med.* **2011**, *41*, 199–220. [CrossRef] [PubMed]
26. Clemente, F.M. The Threats of Small-Sided Soccer Games: A Discussion About Their Differences With the Match External Load Demands and Their Variability Levels. *Strength Cond. J.* **2020**, *42*, 100–105. [CrossRef]
27. Gantois, P.; Piqueras-Sanchiz, F.; Cid, M.J.F.A.; Pino-Ortega, J.; Castillo, D.; Nakamura, F.Y. The effects of different small-sided games configurations on heart rate, rating of perceived exertion, and running demands in professional soccer players. *Eur. J. Sport Sci.* **2023**, *23*, 1214–1222. [CrossRef] [PubMed]
28. Halouani, J.; Chtourou, H.; Gabbett, T.; Chaouachi, A.; Chamari, K. Small-sided games in team sports training: A brief review. *J. Strength Cond. Res.* **2014**, *28*, 3594–3618. [CrossRef] [PubMed]
29. Hulka, K.; Weisser, R.; Belka, J. Effect of the pitch size and presence of goalkeepers on the work load of players during small-sided soccer games. *J. Hum. Kinet.* **2016**, *51*, 175–181. [CrossRef] [PubMed]
30. Koklu, Y.; Sert, O.; Alemdaroglu, U.; Arslan, Y. Comparison of the physiological responses and time-motion characteristics of young soccer players in small-sided games: The effect of goalkeeper. *J. Strength Cond. Res.* **2015**, *29*, 964–971. [CrossRef] [PubMed]
31. Asian-Clemente, J.A.; Rabano-Munoz, A.; Nunez, F.J.; Suarez-Arrones, L. External and internal load during small-sided games in soccer: Use or not floaters. *J. Sports Med. Phys. Fit.* **2022**, *62*, 301–307. [CrossRef] [PubMed]
32. Riboli, A.; Esposito, F.; Coratella, G. Technical and locomotor demands in elite soccer: Manipulating area per player during small-sided games to replicate official match demands. *Biol. Sport* **2023**, *40*, 639–647. [CrossRef] [PubMed]
33. Clemente, F.M.; Praca, G.M.; Bredt, S.; van der Linden, C.M.I.; Serra-Olivares, J. External Load Variations Between Medium- and Large-Sided Soccer Games: Ball Possession Games Vs Regular Games with Small Goals. *J. Hum. Kinet.* **2019**, *70*, 191–198. [CrossRef] [PubMed]
34. Pareja-Blanco, F.; Sanchez-Medina, L.; Suarez-Arrones, L.; Gonzalez-Badillo, J.J. Effects of Velocity Loss During Resistance Training on Performance in Professional Soccer Players. *Int. J. Sports Physiol. Perform.* **2017**, *12*, 512–519. [CrossRef] [PubMed]
35. Koo, T.K.; Li, M.Y. A guideline of selecting and reporting intraclass correlation coefficients for reliability research. *J. Chiropr. Med.* **2016**, *15*, 155–163. [CrossRef] [PubMed]
36. Hopkins, W.; Marshall, S.; Batterham, A.; Hanin, J. Progressive statistics for studies in sports medicine and exercise science. *Med. Sci. Sports Exerc.* **2009**, *41*, 3–13. [CrossRef] [PubMed]

37. Nedelec, M.; McCall, A.; Carling, C.; Legall, F.; Berthoin, S.; Dupont, G. Recovery in soccer: Part I–post-match fatigue and time course of recovery. *Sports Med*. **2012**, *42*, 997–1015. [CrossRef] [PubMed]
38. Dupont, G.; Nedelec, M.; McCall, A.; McCormack, D.; Berthoin, S.; Wisloff, U. Effect of 2 soccer matches in a week on physical performance and injury rate. *Am. J. Sports Med*. **2010**, *38*, 1752–1758. [CrossRef] [PubMed]
39. Ispirlidis, I.; Fatouros, I.G.; Jamurtas, A.Z.; Nikolaidis, M.G.; Michailidis, I.; Douroudos, I.; Margonis, K.; Chatzinikolaou, A.; Kalistratos, E.; Katrabasas, I.; et al. Time-course of changes in inflammatory and performance responses following a soccer game. *Clin. J. Sport Med. Off. J. Can. Acad. Sport Med*. **2008**, *18*, 423–431. [CrossRef] [PubMed]
40. Massamba, A.; Kouzkouz, N.; Geny, B.; Favret, F.; Hureau, T.J. Neuromuscular Fatigue in Hamstring and Quadriceps Muscles after Simulated Soccer Match Play and Typical Training Sessions. *Med. Sci. Sports Exerc*. **2025**, *57*, 984–995. [CrossRef] [PubMed]
41. Harrison, P.W.; James, L.P.; McGuigan, M.R.; Jenkins, D.G.; Kelly, V.G. Resistance Priming to Enhance Neuromuscular Performance in Sport: Evidence, Potential Mechanisms and Directions for Future Research. *Sports Med*. **2019**, *49*, 1499–1514. [CrossRef] [PubMed]
42. Morin, J.-B.; Gimenez, P.; Edouard, P.; Arnal, P.; Jiménez-Reyes, P.; Samozino, P.; Brughelli, M.; Mendiguchia, J. Sprint acceleration mechanics: The major role of hamstrings in horizontal force production. *Front. Physiol*. **2015**, *6*, 404. [CrossRef] [PubMed]
43. Nakamura, F.Y.; Flatt, A.A.; Pereira, L.A.; Ramirez-Campillo, R.; Loturco, I.; Esco, M.R. Ultra-short-term heart rate variability is sensitive to training effects in team sports players. *J. Sports Sci. Med*. **2015**, *14*, 602–605. [PubMed]
44. Edmonds, R.C.; Sinclair, W.H.; Leicht, A.S. Effect of a training week on heart rate variability in elite youth rugby league players. *Int. J. Sports Med*. **2013**, *34*, 1087–1092. [CrossRef] [PubMed]

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